Literature-Implied Status for arXiv:1310.6874

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Status. This note records a literature-implied partial answer to an open problem stated in Daniel Kornlein's *Quantitative results for Halpern iterations of nonexpansive mappings*, arXiv:1310.6874. It is not a new result of the run. The arbitrary uniformly smooth case remains outside the scope of the supporting theorem.

Source problem. Kornlein obtains a rate of metastability for Halpern iterations in uniformly smooth Banach spaces relative to a rate of metastability for the resolvent path (z_t). Near the end of the introduction, after the display describing the form of the Halpern bound, the paper states that it is still an open problem to extract a rate of metastability for (z_t) in the uniformly smooth case, and notes that such a result would include the spaces L_p, $1 < p < \infty$, $p \neq 2$.

Supporting theorem and identification. Kohlenbach and Sipos, *The finitary content of sunny nonexpansive retractions*, arXiv:1812.04940, extract a rate of metastability for the Reich approximating/resolvent curve in Banach spaces which are both uniformly convex and uniformly smooth. Their introduction states that this covers all L^p spaces for $1 < p < \infty$ and gives the first explicit rate of metastability for the Shioji–Takahashi extension of Wittmann's Halpern theorem to this class of spaces.

The paper's Section 7 then applies the extracted resolvent bound to Halpern iterations: it states that the earlier Kohlenbach–Leustean proof-mining analysis was modulo resolvent convergence, and that the new paper is in a position to complete that analysis under the additional hypothesis that X is uniformly convex. The displayed theorem in that section gives, for uniformly convex and uniformly smooth X, a bound $N \leq \Sigma(\varepsilon, \omega_\tau, g, b, \Theta_{b,\eta,\tau,\theta,\alpha,\gamma}, \beta_1, \beta_2, \beta_3)$ such that $\|x_m - x_n\| \leq \varepsilon$ for all $m, n \in [N, N + g(N)]$.

Scope. This answers the source problem for the important uniformly convex-and-uniformly smooth subcase, in particular for all L_p, $1 < p < \infty$. It does not settle the full uniformly smooth case without uniform convexity.

Provenance. This is a literature-implied identification. The supporting paper explicitly cites Kornlein's arXiv:1310.6874 in its bibliography and explicitly presents a Halpern-iteration consequence, but the relation to the exact source sentence is made here by matching Kornlein's missing resolvent-metastability input with the Kohlenbach–Sipos resolvent-metastability theorem.

References.

- D. Kornlein, *Quantitative results for Halpern iterations of nonexpansive mappings*, arXiv:1310.6874; J. Math. Anal. Appl. 428 (2015), no. 2, 1161–1172.

- U. Kohlenbach and A. Sipos, *The finitary content of sunny nonexpansive retractions*, arXiv:1812.04940; Comm. Contemp. Math. 23 (2021), no. 3, 1950093.
- U. Kohlenbach and L. Leustean, *On the computational content of convergence proofs via Banach limits*, Phil. Trans. R. Soc. A 370 (2012), 3449–3463.